



Home energy management system (HEMS): concept, architecture, infrastructure, challenges and energy management schemes

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Abstract

With the rapid advancements in technologies like smart grid, network communication, information infrastructures, bidirectional communication medium's, energy conservation methodologies and diverse techniques, Home area networks (HANs) have undergone a revolutionary change pertaining to various areas of power consumption domains like electricity usage patterns, energy conservation at consumption premises, etc. Under a robust smart grid paradigm, modern home equipped with HEMS contributes significantly towards efficiency improvement, economizing energy usage, reliability, as well as conserving energy for distributed systems. The objective of the paper is to present a comprehensive review covering the various technical and conceptual aspects of efficient power management at home front. A systematic review proposed by several researchers till date in this area is considered. The study focuses on the concepts, technical background, architecture and infrastructure along with various schemes as well as goals including various issues and challenges faced with HEMS systems. The research paper proposes a novel methodology for improvising the home design architecture by incorporating the concept of green building in order to reduce the energy consumption done by a resident at their home front. The concept of HEM systems or SHEMS is not just about proposing new models to save energy, power management, or making energy efficient appliances to be used at home front but also about creating awareness among the consumers and motivating them to participate in the activities related to power conservation in active manner. The study have shown the, impact of various schemes over the peak load reduction with Optimization based residential energy management (OREM) technique showing the reduction in power consumption showing a cut down of about 35% in the overall electricity bill.

Keywords HEMS · HEMS architecture · Infrastructure · Power consumption · Smart grid usage · HEMS working

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1 Introduction

The asymptotic rise in technology has revolutionized the lifestyle of humanity and changed our world into a digital era. With the increased usage of electronic gadgets in our day to day life, the demand for sustainable power supply or electrical energy is always in peak demand. Ranging from general household appliances to individual requirements in diverse areas, i.e. medical centers, electric transportation, factories, information and communication technologies organizations, without electricity supply everything gets disabled. Hence, we may conclude that we are living in a smart gadget's based interconnected and electro-dependent society where electricity is the vital component that keeps these gadgets going [1]. The constant growth in the need for power has raised the cost of electricity per consumption. In other words, the energy cost has been raised after the power generating plants have been deregulated in many areas [2]. The growth in demand for energy is increasing which may anytime exceed the implementation of conventional energy generation systems significantly. This scientific assumption has raised the alarm among the energy sectors in the world which motivated the researchers to propose alternatives like mystification of energy sources such as coal, natural gas oils, fossil fuels, etc. but their use has resulted in negative impact over the environment. This resulted in a shift of focus area of the scientists towards evolving strategies that aimed at saving and developing an alternative source of power generation. The current era of launching new electrical and electronic devices into a market that increases the quality of lifestyle led by people has significantly increased the demand for electrical energy. Moreover, lack of awareness among the consumers towards maintaining a sustainable usage of electrical energy for household consumption, makes it all more challenging on the part of promoters to motivate them towards energy saving at home front. However, launching a model or device has to be promoted appropriately to be enforced, it motivated the scientists to introduce the concept of HEMs. The HEMs allows users to participate in the electric power market enabling the modulation of demand curve according to the specific user profile. This method involves use of intelligent sensors, placed at each point of consumption and software managing the database that stores the users' profile. Generally, the conservation of domestic energy is done in a sporadic and non-autonomous manner. The automation of energy consumption yields optimization benefit in a significant manner. To be precise, a house may be assumed as a connecting junction that facilitates power transmission between the electrical grid and electrical appliances. The HEMS implemented at homes devices such as electrical sensors, relays, data network, and flexible computing platform guarantees an efficient management process. The focus of HEM is towards prioritizing the load consumption concerning the availability of both cost and energy [2].

However, the family circles of today have been advancing towards, in support of the information passing to and fro within the family and circles regarding smart HEMs. Recently, HEM being integrated with concept of providing reviews and controlling methods for residential energy usage has become quite

popular. This concept has influenced advanced countries like USA where the statistics show consumption of 37% of the energy generated [3, 4], Dutch residential consumption shows 24% of energy produced. Apart from practical awareness amongst the residents, scientists have also considered this area as one of the research problems. The authors in article [5] and [6] have proposed a psychological model, illustrating feedback related research question, i.e. how and why feedback is needed. Further numerous models, projects, and scientific studies are carried out focusing towards reducing energy consumption and behavioral changes of consumers.

Before coming to model design aspect for reducing the energy utilization or conducting awareness programs to educate mass regarding the necessity towards energy conservation or how HMS schemes can benefit, it is necessary to understand the technical background, and infrastructural details of the HEMs. In this paper, the goal is towards providing a descriptive overview of HEMS with in-depth information regarding concept, architecture infrastructure and the challenges faced during its implementation and how it can be integrated with other novel concept to attain goals at the home front.

1.1 Organization of paper

The research paper is organized as: Sect. 2 discusses general overview on home energy management system (HEMS). Section 3 illustrates the underlying infrastructure. Section 4 outlines various energy management schemes. Section 5 enlists various challenges faced while implementing HEMS concept in residential homes. Section 6 highlights the real-time practical applications of HEMS. Section 7 mentions the future prospects as well as goals to be attained by HEMS and Sect. 8 concludes the paper.

2 Overview of home energy management systems

HEMS are broad implications of heavy electric power consumption at home front that has caught the attention of current generation scientists. The luxurious lifestyle led by the current generation where day-to-day chores are carried out by electrical appliances and new innovative devices being launched in the market every now and then, demands for resilient energy flow surpasses the amount being generated. This section highlights the objectives, measures and research work undertaken towards solving this problem. In addition to this the concept of HEMS, architecture and its functionalities are discussed in detail.

2.1 History of HEMS

The concept of HEMS was started in the year 1979 where a functioning Energy management system was conceptualized based upon the article “Solar energy management system” written by R. Moen [7]. After that, the system proposed underwent

performance improvement specifically with the introduction of PCs in 1980 [8]. In the year 1982, Optimization Algorithm was designed for managing power with an aim to reduce electrical cost by cutting down demand as well as usage time [9]. Next, Novel strategies were devised for controlling and management of electronic machines, using a home automation communication network system [10]. HEM system designed specifically for home-based applications based on residential automated communication mode, where customer tracking and missing object location could be obtained using technologies like radio frequency, video technology, ultra-sonic sensors which was proposed simultaneously [11]. In 2003, [12] power-controlled system was inducted for controlling energy input of 20 residents in Japan by use of gateways from every home. The gateway served as an automation collector regarding power consumption, information controls, etc. This was improvised with network architecture, power line communication which serves similar purpose as an energy management controller integrated with PC's for controlling and monitoring electrical appliances [13]. Further HEM schemes involving Intelligent Algorithms were designed [14], which was used as a tracking device for the Undergoing activities of the single occupant as well as locating the resident's status in similar environmental conditions. Later on, the resulting outcomes proved an increase in performance level, the comfort of occupants as well as reduced energy consumption. In 2006, a patent was filed by whirlpool corporation describing HEM controller, which could manage the energy usage in home-based devices [15]. This technology launched EMS which received information regarding the off-peak time slots for the day. HEMS enhanced it with infra-red remote (IR) controls designed for controlling electric lights and power sockets within houses [16], but it failed to cover the whole distance from outlets to central controller. In 2012 with the advent of intelligent HEMS with DR began the trend of advanced technology in HEMS to lower power usage and electric bills [17], for home based electrical appliances such as A/C, electricity driven motor vehicle, etc.

During the same time smart HEM was conceived, integrated with controller based on event-driven binary linear optimization, for providing optimum electric energy within the residential areas [18]. Rest of the work used Advanced Optimization Algorithms to schedule HEM which considered the availability of large storage system, renewable resource, and variable electricity cost to lessen power requirement for a functioning house [19]. Similar novel algorithms regarding lower electric consumption or smart electricity management were proposed with various energy management models, such as HEM with renewable energy resource [20], adaptable smart HEMS by accounting the effect of power usage pattern of network with respect to dynamic variable such as weather, electricity tariff price, etc. Studies were also been conducted regarding implementing smart HEMS via real-time energy control [21], a fuzzy controller for determining charging/discharging battery power, and energy scheduling and consumption using rolling optimization [22]. HEMS systems designed using intelligent lookup tables based on associative neural network for determining optimal energy efficiency, etc. [23, 24] that may help in minimizing energy consumption, HEMS framework was proposed using artificial bee colony as an optimization technique by scheduling home appliance [25]. Apart from diverse scientific research proposal's, few companies have also undertaken the initiative

towards designing intelligent HEMS for controlling energy utilization, e.g. General Electrical Co. for designing home appliances using smartphones [26, 27], Honda, US for implementing hardware system that can possibly control, monitor as well as optimize, electrical generation and appliance with use of, at home-front [28]. The home automation system designed consists of Solar PV, batteries for energy storage that may be used during the peak hours of the day.

2.2 Concept of HEMS

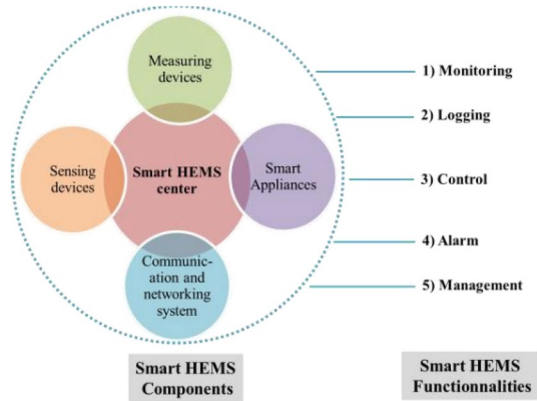
Beneath the umbrella of grid architecture, AMI devices have activated a dependable communication mode connecting both power utilities and residential consumers [29]. This communication channel paved the way for an opportunity to include the concepts of economic incentives conceptualized for a smart home for managing the demand-side resource by switching to and fro their energy consumption during peak-load hours of the day as a method to shed load for reduced electricity bills. Such economizing methods involve savings in electric bill, improved smart home appliances, for residents and adoption of various energy conservation methods, etc. [30]. Therefore, smart HEMS may be defined as a home that offers energy management services for efficient monitoring and management of electricity generation, power conservation, as well as energy storage methods designed within the smart home [31, 32]. Moreover, it also includes the communication and sensing services that facilitates the statistical survey of energy drawn from various residents. The different operational mode of the various smart home devices along with their real time monitoring can be obtained by using device such as PC or mobile phones. Apart from the house hold consumption status HEMS also provides facilities for distributed energy resources (DERs) and HESS [33, 34].

2.3 Functionalities of HEMS

To participate in power conservation and respond to the growth in demand, HEMS is required to be more adaptive towards managing as well as controlling electronic devices or even renewable energy resources [35]. The proactive control operations like real-time information regarding units of power consumed, its cost in homes can be offered to the purchasers by HEMS. The residents may opt through human-machine interactive platform to specify the functioning slots of different domestic machines to improvise their energy usage ability [36]. Specific features of a smart HEMS center consist of five major functioning modules as shown in Fig. 1.

- **Monitoring**—the process of monitoring makes real-time information regarding energy usage pattern accessible. It facilitates the energy utilization and shifts user focus towards electricity conservation. It may offer visual displaying of services for several of its operational modes along with energy status of each household devices.
- **Logging**—logging is the process of collecting and saving the data information pertaining to the unit of electricity consumed by each appliance, generated out

Fig. 1 Functionalities of HEMS
[102]



of DERs and energy conservation states. This functionality includes analyzing demand response for real-time prices.

- **Control**—control factor can be segregated into direct and remote control. Direct control may be defined as control exercised on both electronic device and the control system, whereas the remote controls signifies the customers hold towards accessing, monitoring as well as controlling the consumption patterns of their electronic devices through various other systems and devices, e.g. PCs or smart-phone.
- **Management**—it is one of the primary function of HEMS which improves both optimization as well as efficiency in the electrical power utilization within the smart home. It can also cover operations that included renewable energy system management service, energy storage management service, home appliance management service, and Plug-in EV and battery management service.
- **Alarm**—here alarms are generated as well as passed on to the smart HEMS center which contains information regarding fault locations, types, etc.

2.4 HEMS architecture

A home energy management system (HEMS) [37–39] is defined as a system that inculcates sensors within home devices, via home networks. The HEMS in majority are developed with a purpose of controlling power utilization, bringing improvement in the performance level of a smart grid, optimizing demands, enabling devices in the residential users, etc. Here HEMS play the role of a modern energy meter bringing an evolution in the form of low-consumption appliances [40, 41]. To be precise, HEMS smartly monitors and adjusts power consumption by smart meters, smart devices, appliances and smart plugs, resulting in efficient power and better management [42–44]. The architectures, proposed by researchers in different articles [45, 46] may be categorized under various heads, considering criteria related to the structure of monitoring, management, and distribution and communication capabilities.



Fig. 4 Components of smart home energy management system

shifting, is considered to offer the solutions to network operators that have boosted the drive for robust as well as intelligent SHEMS.

The major components of SHEMS are [51, 52]:

- The sensing device
- Enabling ICT
- Smart appliance
- Energy management system

Figure 4 shows the various components participating into making a home, smart home.

- **The sensing device:** the devices are used in a home front which contributes to the SHEMS concept deals with sensing current, voltage, temperature, motion, light, and occupancy. Various other sensors that have been specially designed for health, safety, and security reasons are smoke, epilepsy, sensors, etc. They detect the presence of desired parameters at various locations and send its signal to a centralized system. Considering these parameters, the smart appliance is possibly monitored, controlled or scheduled to function at the specified periods of time.
- **Enabling ICT:** the ICT may be treated as the linking pin that connects together the sensors, meters and devices with monitoring or the control unit. Here both wired as well as the wireless communicating methods are devised in order to intricate various domestic appliances. Wi-Fi, networking, Home plug, as well as Z-Wave are certain leading technologies that fascinate the Home Area Networks [53, 54].
- **Smart appliances:** a smart appliance system offers the residents an awareness regarding their energy usage, the action of enabling energy efficient and eco-

friendly. They are primarily the domestic appliances that are equipped with intelligence as well as communication system which facilitates remote mode of controlling as well as monitoring.

- Energy management system: currently the system lacks a fit for all solution since various developers currently concentrate over different aspects. The energy management principles, software platforms and the embedded intelligence in SHEMS varies from manufacturer to manufacturer [55, 56].

Few basic functionalities of HEMS are as mentioned below:

1. It provides an informative overview regarding the graphic forms of energy usage data.
2. Its few advanced functions typically consist of information, automation, monitoring etc.
3. It also includes chances of the forecasting as well as scheduling of loads and local generations at the house hold levels.
4. The automated operations here offer the customer selection in order to determine the priorities with the home equipment and local generation of operations.

2.6 Smart home renewable energy management system

A smart home energy management system (SHREMS) is a system capable of inter-changing commands between households and energy providers leading to optimization the energy consumptions. This joint effort between energy shareholders result in reduction of consumer electricity bill, efficient management of the peak loads by the electrical utilities, etc. [5]. Smart grid is considered as one of emerging R&D concepts that integrates both traditional electrical grid as well as recent development of information and telecommunications technologies in order to improve the efficiency of power generations, transmissions, distributions and consumptions systems [55]. One of the key features of the smart grid is the integration of renewable and storage energy resources at the consumption side [54]. It also enables consumers and utilities to communicate with each other in order to share the responsibility of managing the power flow and consumptions [56]. A single chip microcontroller is used to multiplex the three power sources to supply the house with its required power based on communication between the utility and the house owner. The communication protocol, energy flow, demand-response and billing system hardware and software are developed using a home gate and utility server. The home gateway (H-Gateway) is a single chip embedded system integrated with GSM modem and it is installed at the consumer premises. The utility server (U-Server) is a high-end PC and it is installed at the utility headquarter. Consumer and Utility can manage the energy follow and consumption by exchange of message between the H-Gateway and U-Server via the GSM modem [57].

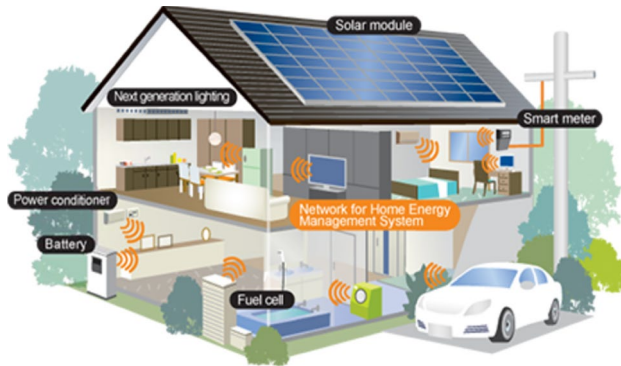


Fig. 5 Network for HEMS

3 HEMS infrastructures

The HEMS deployed on domestic fronts typically consists of smart HEMS center, smart meters, communication and networking system, HESSs and other smart devices [53, 54]. These devices aid electronic home appliances in controlling the electricity requirement. HEMS role is to access, monitor, and control as well as optimize overall performance of different DERs, EVs, household electronic devices, etc. [55, 56]. They are optimistic towards supporting fully integrated smart home electrical appliances and a smart home supporting interaction in both ways, among consumers and electrical power utilities. Various attributes of smart grid include low-cost, providing unique services, adaptability, advanced smart power technology, user friendliness which may also be made operational via HEMS [57–59].

3.1 Communication and networking system

Various HEMS systems are conceptualized, designed and proposed and built on various networking and communication-based schemes along with specific hardware implementation [45–51] as shown in Fig. 5. Few e.g. are power line communication [40], ZigBee [46, 52], BACnet [47], Bluetooth [48] and human–machine interface systems [52, 53]. Rigorous studies have been conducted on the communication and networking areas in a HEMs. According to article [36], the hybridization of both smart meter and power line communication as a unit was able to facilitate concept like remote accessing, planning, and power conservation of domestic electronic devices at home front. Researchers in article [39], proposed a novel Smart HEMS basing upon an IEEE 802.15.4 and ZigBee, that groups together for allocating various network related activities to fitting components in house [45]. The support of active sensor networks, in integration with sensors and actuators, supports HEMS for diversified environmental sensing information as well as into controlling the variety of household devices. Table 1 highlights the various communication accessories and devices required in HEMS. Researchers of article [51] have highlighted

Table 1 Types of communicating modes supported in HEMS

HAN communication and network technology	Wired Mode of communication	Wireless radio frequency	
Mode of communication	Wired electrical wirings, telephone line, optical fiber	Battery operated 0-energy harvesting	Battery free-energy harvesting
Supporting devices	Home Plug, Ethernet, X10*, Insteon	Z wave, Wi-Fi, ONE-NET,	EnOcean
Supporting protocols	ITU G.hn (IPv4, IPv6)	6LoWPAN (IPv6)	

the need to build automation systems and communication infrastructures which is basically designed on BACnet.

Considering the capability of embedded Bluetooth technique inside a communicational device and home-based machines, researchers [49, 58, 60] formulated a strategy of designing a fully Bluetooth integrated covered home network. In addition to this, researchers [51, 61, 62] have established a human-machine interface system for HEMS. ZigBee can be defined as a wireless communication technology, technically proliferated in recent trends. It uses 2.4G–2.4835 GHz frequency band. In areas of low energy usage and longevity, it serves as a low power cost requiring technology holding large quantity of features of the network node, making it quite apt for industrial applications [55, 63, 64].

3.2 Smart meters

Smart meters may be defined as advanced energy meters which can quantify the power drawn for a user along with providing extra details regarding power utilities based upon a bi-way communication method [64, 65]. This helps the consumers towards malign optimal decision for scheduling electricity usage for residents along with the actions of DER's and HESS. Smart meters merged with computer technology evolves as a recent and efficient technique for data acquisition, processing as well as management of advanced metered equipment that contributes as major foundation [66, 67]. Smart meters being the background of the smart electricity, its operational design takes into account their current factual as well as future developments, e.g. capacity of communication channels, accessing DER, etc.

The primary functions of the smart meter include [52, 53, 65]

- Measurement of both the multi-period and multi-mode power rates for active as well as reactive energy consumption.
- Allowing bi-way communication—data and information can be sent as well as accepted, e.g. real-time information query, up gradation programs for electric meters and settings, etc.
- Responding needs for achieving smart load shedding, working with other smart meter and smart interactive terminals when other devices fail to work due to problem in the main power grid.

- Data collection using various smart gas meters.

3.3 Smart HEMS center

A smart HEMS center can be compared to the working brain controlling the whole of smart electricity home and implements the smart energy handling capability [66–68]. A smart HEMS center may be placed within a user's residence as domestic appliances. Its main functions include [69, 70]:

1. Receiving massive quantity of data send by smart meters, central control panel, and real-time display. The consumer sends controlling instructions to every household equipment's that results into an automated demand response.
2. It provides a friendly human–machine interaction platform that supports the user's real-time browsing activity, online monitoring, task setting and other functionalities to administer the use of electricity and functions that can administer the electric usage.
3. A smart HEMS center can set water, electricity, gas, etc. with a colossal scalability.
4. Integration of DERs, energy storing equipment and electricity regulating devices for EVs, forecasting distributed generations for the optimal control of DERs, etc. [71].

3.4 Home appliances

The variety of household machines and energy storage devices can be thoroughly analyzed as well as modeled basing upon its characteristics, usage pattern, etc. [72, 73]. For implementation purpose, an optimally coordinated trendy home-based devices can be categorized as:

1. Non-schedulable domestic devices, e.g., freezer, printing machines, micro ovens, TV, hairdryer.
2. Schedulable home-based machines that can be set or reset for optimal operation and may be switched on/off during any time, e.g., washing machine, ACs, iron, geysers, EVs, i.e., those machines that can fulfill a given task in the absence of manual control. [74, 75].

The extent of convenience of customer is considerably affected with the performance shown by a non-schedulable appliance, where the service is most probably on time. Schedulable appliances may be further grouped as 'interruptible' and 'non-interruptible' denoting the continuity of operating time. The interruptible appliances are normally more schedulable as compared to an uninterruptible scheduling. The non-interruptible appliances are a constraint to fixed operation period termed as the 'hold-time' [76, 77]. As a separate load, an EV gains importance as well as extensiveness. Experts consider, an increase in the quantities of EVs can be raised in future, which would contribute in reducing air pollution

and greenhouse gas emissions [78] to huge measure. As EVs are charged or discharged while in connection to a grid, the rise in practical service may be obtained through power grid [53]. Vehicle-to-grid is a novel method introduced to show the charge stored in an EV battery which can possibly be transferred to a power grid [54–56]. Smart HEMS, consists of EVs that are able to handle the peak energy, indicating power supply when peak periods and consumer's usage during off-peak periods.

3.5 Machine to machine communication

The concept of machine to machine communication has evolved as a cutting-edge technology for communication with respect to future generation. It has undergone rapid development while inspiring multiple set of applications. The main concept of M2M communication is to establish the interconnection network among M2M components. The interconnectivity of machines has gathered huge attention over the years. Its supporting features are like [75, 76, 78].

- Being remotely controlled;
- Low cost;
- Scalable;
- Reliable technology, have made them quite popular among customers. Their application areas consist of public safety, energy management, transportation.

The M2M network is primarily composed of following 5 elements:

- It is usually embedded within a smart electrical device in order to reply request as well as transmit the data.
- The M2M gateway which supports into establishing the connectivity among various M2M components and communication network.
- The M2M server here functions here as middleware layer in order to move data through a various application service.
- M2M application can be used to monitor as well as control status, power consumed and demanded, etc.
- M2M Server are typically designed for establishing and supporting connection among various modules.

The M2M area networks as shown in Fig. 6 support connectivity between M2M gateway as well as M2M server. The mentioned elements build up the three domains of M2M component working over the device domain whereas the M2M area network and the gateway in network and the gateway in network domain whereas the M2M server and communication network come under application domain.

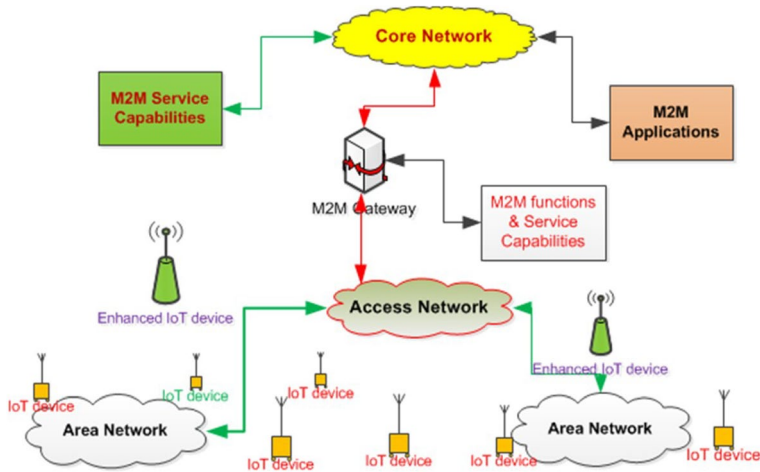


Fig. 6 Network communication for machine to machine [103]

Table 2 Comparative analysis on various HEMS schemes

Name of HEM scheme	Pricing technique used	Objective	Methodology	Area of coverage	Peak load reduction	Bill reduction
OREM	ToU	Reduction in electricity consumption bill	Linear programming based optimization	HAN	N/A	35%
HACS4EM	ToU	Reduction in peak to average ratio, better peak load management	Scheduling of Home appliance	HAN	21%	16–25%
RLC	RTP	PAR and cost minimization	Linear programming based optimization	HAN	22%	24%
OLM	RTP	Decrease in electricity bill and consumption	Heuristic optimization technique	HAN	N/A	8–22%
DST	ToU	Scheduling DES	Particles swarm optimization (PSO)	HAN	N/A	16–25%
i-HEM	ToU	Cost maintenance	Interactive Load shifting (ILS)	HAN	40%	30%

4 Energy management scheme

In home energy management systems, generally 30% of the total power consumption occurs during on-peak hours of the day. With implementation of HEM program the load was found to be reduced up to 5%, which can be considered as a positive contribution towards the reduction in electricity bills, GHG emission, energy consumption, etc. [73–75]. In this section, various proposed management schemes by different researchers with regard to HEMS are highlighted. Table 2 shows a comparative analysis of the various management schemes proposed.

4.1 Optimization based residential energy management (OREM)

Erol-Kantarci and Mouftah [76], have proposed a linear programming (LP) based model conceptualized to reduce electric consumption at domestic front. This method divides a day into time slots having equal intervals where electric rates varies for every unit, quite similar to that of ToU scheme. In this model, the domestic electrical machines are scheduled to be operated upon at appropriate time slots, to lessen power expenses. The consumers in LP model give schedule detail as input, where the model provides extremely efficient and optimized output schedules. To achieve minimum electricity unit, price the LP model schedules the appliance operating time in the minimum rate time slot. This technique incurs huge delays into the appliance cycle that creates chaos during low-price time slot. To overcome the problem, authors have introduced the concept of maximum possible delay for every appliance.

Advantage: the strategy proposes efficient management of the energy consumption (OREM) by scheduling home appliances at the time slot.

Disadvantage: the technique of scheduling introduces additional amount of delays to the appliance cycle.

4.2 In-home energy management (i-HEM)

Han et al. [77] demonstrated a novel home management algorithm designed to coordinate as well as interact among trendy devices, central EMU and WSHANs. This communication mode is built upon the Zigbee protocol for wireless communication. In contrast, i-HEM works on real-time pricing [76]. Here the user is provided with the flexibility to swift within any appliance, any point, any-time, without considering the consequences. The i-HEM algorithm is about suggesting perfect time for consumer, where they can accept overriding the suggested slot.

Advantages:

1. Here consumer demands are processed on a near real time basis, i.e. the consumer has the flexibility of turning on any appliance at any moment on the clock irrespective of the peak hours.
2. Reduces carbon emission and energy consumption cost.

Disadvantage:

In absence of energy management program, 30% of the load takes place during peak hours for the i-HEM application.

4.3 Home appliance coordination scheme for energy management

Mahamood et al. [78] highlighted the need of HEM, being designed on coordination and interaction among appliances, same as i-HEM. Here Sensors are used with a purpose to monitor attributes of all the working household appliances. The home-based devices are integrated with sensors in combination with AMI that helps in monitoring real-time power usage. This scheme is based on ToU pricing mode. It also addresses the function of standby devices where, the standby power denotes, power drawn by appliance while operating in nonfunctioning or switched off mode [79]. Pedrasa et al. [80], have mentioned the electricity usage of appliances to be all-total 10% of the electricity [81]. The primary aim of Appliance Coordination (ACORD) algorithm is about shifting the appliance from on-peak loads to off-peak hour. The consumers, interest in the energy management system may improve the efficiency of the system. With the increase in some consumers requesting an addition in energy, system significantly reduces the electricity bills [81, 82].

Advantages:

- Increased savings.
- Efficient peak load management.
- Reduction in peak to average ratio.
- Aims at low cost and low power consumption.

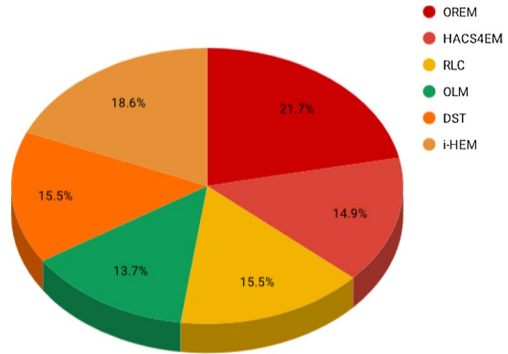
Disadvantages:

- Lack appliance coordination.
- It needs efficient electricity consumption in Smart Grid.
- Packet size here effects the delivery ratio.

4.4 Decision support tool (DST)

The primary role of DST is towards enhancing consumer's convenience by taking smart decision while coordinating electronic appliance. The HEMS result values shows raise when DES is simultaneously coordinated along with the devices. The research contribution in [83] uses an enhanced PSO solver for this purpose.

Fig. 7 Shows the reduction in energy consumption for various HEMS schemes



Coordination, as well as scaling, is generally conducted on two basis, namely when DES is independently scheduled or when it is cooperating with each other. DST is a combination of DES scheduling along with appliances coordination to maximize the system output. The net result of this algorithm is a reduced electricity bill by 16–25%.

Advantage:

- Integrating DER coordination in parallel with appliances enhances the Energy Management Program and its functionalities.

Disadvantage:

- It requires scheduling algorithm in order to be beneficial to consumer.
- Requires proper coordination with devices to provide result.

4.5 Optimum load management (OLM) strategy

In article [81], an objective function is defined by user for purpose of providing both forecast as well as schedules of appliance activities. Here various options may be considered before forecasting such as solar, wind energy, per unit price of electricity at the different time slot of a day. Following the objective function as proposed in article [104], where, y is the difference function among the amount user has paid and the cost of obtaining the required energy from the grid.

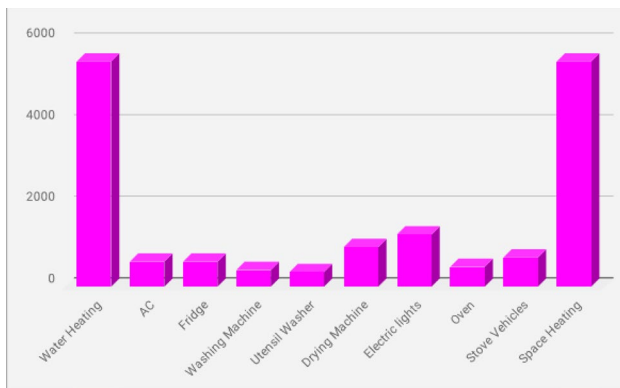
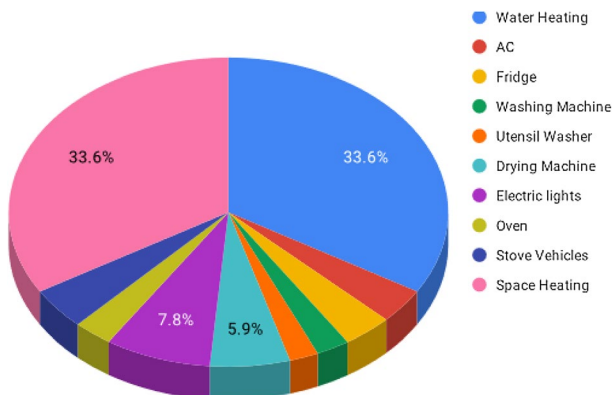
Advantage: the advantage of energy management program may be increased if DER coordination is adopted in parallel with appliance coordination

Disadvantage: it requires scheduling algorithm in order to be beneficial to the consumers. The bill reduction is only 16–25%.

Table 2 shows a comparative analysis performed on various HEMS schemes available to reduce energy consumption. The rate of reduction was found to be more promising in OREM based HEMS scheme proving it to be more suitable as an electricity saving measure. [78–84]. Figure 7 shows the rate of energy reduction achieved by various HEMS schemes.

Table 3 Annual energy consumption in Canada, (Year 2017) [85, 86]

Electronics appliances	Annual energy consumption
Water heating	5.5 MWh
AC	0.6 MWh
Fridge	597 kWh
Washing machine	400 kWh
Utensils washer	350 kWh
Drying machine	964 kWh
Electric lights	1277 kWh
Oven/	470 kWh
Stove vehicles	717 kWh
Space heating	5.5 MWh

**Fig. 8** Annual energy consumption in Canada**Fig. 9** Energy drawn by various electronic household appliances

5 Challenges of home energy management systems

5.1 Diversity of residential electrical devices

Various electronic appliances used at home front operates at different energy level with respect to its underlying machine. A survey was conducted at Canada in year, 2017 to know the energy consumed by various household machines which would help them in designing or framing appropriate HEMS scheme as shown in Table 3.

As observe from above Figs. 8 and 9, there are variety of electronics and electrical devices in every residence that are remarkably diverse with respect to ownership rates as stated in Table 3. E.g. 45.6% of Canadian houses do not own a portable stereo, whereas 42.3% have one, 97.1% have 2 and 2.3% have 3 or even more. Apart from this the variations in both power consumption/production and quantity of use of various devices as shown in article [85, 86]. The features contributing to this diversity and energy drawn patterns of electronic appliances include dwelling attribute, quality lifestyle of people, their financial status, their occupation. This also includes electronic devices such as storage units, electric vehicles that go beyond the mere home appliances pertaining to household. This makes HEMs necessary for adapting various architectures, so as to coordinate along with the schedules of various unique devices.

Electronic appliances like space heaters, hot water geysers, ACs, fridge, plug-in hybrid electric vehicles (PHEV)/electric vehicles (EV), washing machines, utensils washers, renewable energy generation (PV/Wind), dryers, electric lights and oven, stake their positions at the top of the list with water heating and space heating showing the maximum as shown in Figs. 8, 9. This may be reasoned as the fact that energy consumption of these devices is found to be coinciding with most significant contributor's electric loads for residents.

This energy consumption can be readily reduced by integrating the concept of green building in HEMS, where major focus is based on utilization of natural resources as a replacement of electrical appliances while designing homes. The water heating electronic machines can be replaced by solar panels being supported at the rooftop of the building which may contribute into heating the water present in reservoirs reducing the need to use geysers for providing hot water.

5.2 Multi-objective nature of the residential setting

As suggested by the existing literature we can claim that the aim of HEM is to effectively manage the energy consumption which includes attaining a balance between the criteria's such as cost, wellbeing, emission, consumption etc. [87–89].

- **Cost:** this covers the expenses pertaining to energy management. Though energy consumption dominates, it also includes devices, startup costs, rewards, penalties, carbon tax etc. [90, 91].

- Well-being: it signifies the standard of lifestyle and is a vital objective to be maintained despite managing energy consumption. Any losses with regards to class of service rendered by energy delivery results in disruption incurred upon the consumer [90, 92].
- Load profiling: this objective deals with evaluating the significance of load profiling, reduction in peak demand for a service, lessening the grid dependence [93, 94].
- Emissions: it refers to the green-house gas emission that is conjoined to electricity usage. These are attached directly with the electric used based upon intensity of the grid emission, electricity consumed, etc. the emission can be measured by assuming the indirect impact of a specific action. E.g. Semi charged PHEV, controlled use of micro combined heating or power units (μ -CHP), etc. [95].

5.3 The uncertainty of future consumption

Along with diversification within various devices and goals to be attained, there is a major issue of uncertainty regarding houses which include PV, wind power production, people's occupation, behavior and patterns of energy usage, atmospheric conditions, etc. The responsibility of devices scheduling within household often inculcates the forecasting of these uncertainties. The forecasting techniques like NN (Neural Network), SVM (Support Vector Machine), fuzzy can be included. The accuracy level of forecasting strongly depends the techniques used and methods of error measurement [96, 97]. Few typical contributions include an attempt towards forecasting 24 h wind speed, Solar Power, water needs, occupant's behavior, etc. Many works on HEMS can be developed which focuses on scheduling and forecasts which may be provided through external source and can be addressed by different modules of HEMs.

5.4 Infrastructure challenges

To enable the HEMs for active participation, it needs specified data as well as information, and the capacity to send the appropriate signals designated to specified recipient (e.g. utility and loads). Here the modern electricity grid carries less functionalities regarding features such as communication. Hence, a bright powered background called smart grid" is employed that can facilitate the effective manipulation of HEMs with addition of communication layer to the grid [98]. Few communication metrics as highlighted in the article includes Wi-Fi, WiMAX, 3G, DSL, Ethernet, optic fiber, PLC, etc. Within a home front both the HEMs and the electronic appliances communicate through control and information signals for optimizing the consumption as well as the production schedule. The various communication levels have been conceived in a home area network. E.g. are Home plug (IEEE P1901), Ethernet (IEEE 802.3) $\times 10$ ($\times 10$ standards), Einstein ($\times 10$ standards), etc. [95, 96]. Hence it may be claimed that HEM

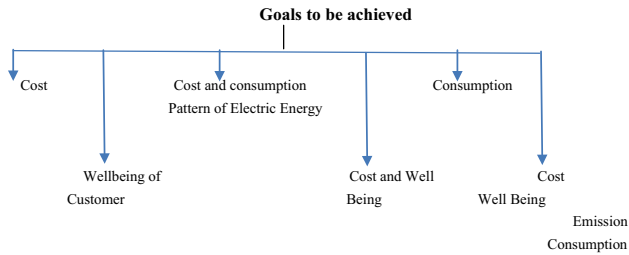


Fig. 10 Future goals to be attained by HEMS

appliances are required to announce major important information, requirements with regards to its states and energy for functioning effectively and for undertaking action based upon the signal it receives from HEMS. The electronic devices that are equipped with such facilities are designated as being smart device or appliance. Hence, we can assume from literature that HEMS have direct control over the devices but may not be needed in areas where smart devices are controlled centrally or areas where SHEMS relays necessary information for smart machine for creating its independent schedules [96].

6 Real time practical applications of HEMS

Few real-time practical applications of HEMS are as follows:

- **HVAC control:** meant for peak cutting or peak levelling control for heating, ventilation, and air condition which works in accord with HEMS to cut down the energy consumption. At any residence, a thermostat may be considered as the HVAC control that manages the operation of the furnace or air conditioner, and the blower that distributes heated or cooled air through out.
- **Energy saving:** HEMS is used for the purpose of thermal comfort and energy savings in Japan residential building currently. It is quite popular among the residents with various schemes and technology promising to reduce the energy consumption at home front.
- **Novel optimized HEMS controller:** any novel optimization methodology can be incorporated among the smart grid providing an efficient demand side Management system. These optimized controllers shows a positive result when implemented in residential premises encouraging it all the more.
- **Performance analysis:** the HEMS designed can be used for performance analysis of various devices along with various algorithms. The comparative analysis helps in choosing the best methodology to be adopted.
- **Promoting energy conservation and maintenances:** energy conservation and maintenance of indoor comfort can be promoted by various workshops and awareness programs conducted for the same. Apart from proposing various effec-

tive technology to save energy consumption, it important to create an awareness among the residents to stop wastage of energy and adopting various schemes at their home to get a reasonable cut in expenses over electricity consumption.

- Integration with other programs: HEMS concept can be integrated with various green building ideas in order to save energy consumption at various residential level via utilizing natural resources to its most.

7 Goals to be attained by HEMS and future prospects

From the literature, the various HEMS design and implementation schemes aim at attaining the following goals listed in Fig. 10.

The critical needs of a comprehensive HEMS originate from research article and are afterwards refined for being used in the comparative analysis presented as under: Implementing control and feedback needs internet connection for providing data's and information into an online platform in order to access useful information.

- Requirements for a productive feedback—monitoring the energy consumption is a basic prerequisite but with the extension disaggregated monitoring gave way to increase in reductions. The concept of disaggregated monitoring brings about need to measure the energy use of power-hungry loads separately [97, 98].
- Offering context to provide actionable suggestion that gives way to further reductions. This goal considers the environmental factor, e.g. temperature, humidity, etc. that influences the amount of energy required during a day [99–101].
- Motivating consumers into building good energy usage habits.
- Regular updating of feedback: the information regarding consumption must be as accurate with respect to time as well as device specifications. [98, 100].
- The system cost should not surpass the accumulated savings while operating since it lowers down the financial incentives associated to HEMs.

Apart from attainment of these goals researchers may also incorporate optimizations techniques or AI/IoT [103] techniques for designing HEMS or SHEMS. The novel concept of green building may also be used in order to replace electrical and electronic devices with natural resources where ever possible which may result in reduction of over power consumption.

8 Conclusion

This article first presents a comprehensive illustration of what HEMS is all about and how it has gained popularity with its issue and challenges. It provides a detailed study on the current state of art, contributions by various researchers on HEMs done so far. Here we discuss architecture, and infrastructure in detail. Various models pertaining to smart technologies, mechanisms, and schemes conceived and designed

with respect to HEMS are presented. Artificial intelligent techniques for developing a HEMS schedule controller based upon rules, ANN, FLC, and ANFIS systems have been compared. The article explains the mathematical models and heuristic optimization techniques integrated into HEMS for developing an optimal schedule and minimizing the energy usage with shift operations of home appliances into off-peak zones without causing an impact on customer comfort. Finally, we can say that the futuristic plan regarding HEMS are in support of self-learning AI techniques which may replace the user position and access in system settings supporting the venture of HEMs and SHEMs into different level of power and electricity management altogether.

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